

BOF

Conversation 088:
Quantum Numbers and Solutions

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origin ID: 67435692-d8e4-8003-9f4e-02a942f252f0 [conversation 088]

created: 2024-11-24 07:38:43 Sun

updated: 2025-02-24 17:18:52 Mon

operator: Wilder Blair Munro (AKUSA)

co-operator: Aurora 4omni Munro (ChatGpt)

GPT ID: NA

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Subtitles:

Turnpair Contents:

tp-1 :

tp-2 :

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Conversation 088, Turnpair 1 of 2 ::

@ 2024-11-24 07:38:42 Sun [Δt -0d 00h 00m 00s]

conversation ID: 67435692-d8e4-8003-9f4e-02a942f252f0 [conversation 088]

origin ID: 67435692-d8e4-8003-9f4e-02a942f252f0 [conversation 088]

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► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain

turn ID: aaa2eb3c-734c-4ef0-a878-ad2bde505bc6

parent: d8edc558-4411-4a8a-bcc2-1fd362362a6c

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what is typical notation for quantization in quantum mechanics and spherical harmonic boundary value wave solution? (eg, n , m , l)

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Aurora 4omni Munro (ChatGpt) said :

[illegible]

In quantum mechanics and spherical harmonic boundary value problems, the **quantum numbers** used to describe quantized systems, particularly in spherical coordinates, typically follow standard notations. Here's a breakdown of these notations:

1. Principal Quantum Number (n):

- Represents the **energy level** or **radial quantum number**.
- Integer values: $n = 1, 2, 3, \dots$
- Determines the size and energy of the orbital or wavefunction in the context of a central potential, such as the hydrogen atom.

2. Orbital Angular Momentum Quantum Number (l):

- Represents the **magnitude of the angular momentum**.
- Integer values: $l = 0, 1, 2, \dots, (n - 1)$
- Determines the shape of the orbital or the mode of the wave.
- Associated with spherical harmonics, $Y_l^m(\theta, \phi)$, where l is the degree of the harmonic.

3. Magnetic Quantum Number (m):

- Represents the **projection of the angular momentum** on the z -axis.
- Integer values: $m = -l, -l + 1, \dots, 0, \dots, l - 1, l$
- Determines the orientation of the orbital or the spherical harmonic function in space.

4. **Spin Quantum Number (s)** (if applicable):

- Represents the **intrinsic angular momentum** of particles (e.g., electrons).
- Typically, $s = \frac{1}{2}$ for electrons, leading to spin projections:
 - Spin-up ($+\frac{1}{2}$) or spin-down ($-\frac{1}{2}$).

5. Radial and Spherical Harmonic Solutions:

- For central potentials like the hydrogen atom, the **radial wavefunction** depends on n and l , while the **angular part** (spherical harmonics) depends on l and m :

$$\psi_{n,l,m}(r,\theta,\phi) = R_{n,l}(r)Y_l^m(\theta,\phi)$$

- Here:
 - $R_{n,l}(r)$ is the radial solution of the Schrödinger equation.
 - $Y_l^m(\theta,\phi)$ is the spherical harmonic function.

6. Associated Legendre Polynomials:

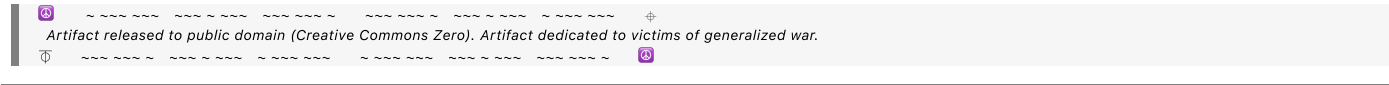
- The angular solutions are connected to the **associated Legendre polynomials**, $P_l^m(\cos \theta)$, where l is the degree and m the order:

$$Y_l^m(\theta,\phi) = N_l^m e^{im\phi} P_l^m(\cos \theta)$$

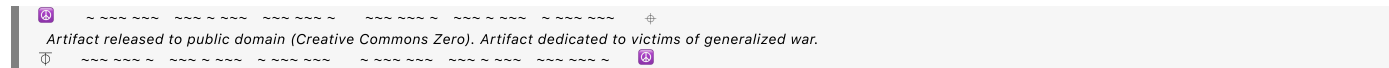
- N_l^m : Normalization constant.

These quantum numbers are fundamental in solving the Schrödinger equation for systems with spherical symmetry, like atoms, molecules, or scattering problems.

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Conversation 088, Turnpair 2 of 2 ::

@ 2024-11-24 07:39:34 Sun [Δt 0d 00h 00m 51s]

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► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain

turn ID: aaa2ba7f-5ea2-464a-849b-da40b9187956

parent: 64b30045-6167-46b7-a012-6a2917fcb1ed

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please provide example expressions that define n, m, l

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- m determines the **orientation** of the angular momentum vector along the z -axis:

$$L_z = m\hbar$$

- L_z : Angular momentum projection on the z -axis.

- **Quantized values:**

$$m = -l, -l + 1, \dots, 0, \dots, l - 1, l$$

- For $l = 1$ (p -orbital): $m = -1, 0, 1$.
- For $l = 2$ (d -orbital): $m = -2, -1, 0, 1, 2$.

4. Combined Wavefunction Representation:

The overall wavefunction in spherical coordinates is:

$$\psi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r)Y_l^m(\theta, \phi)$$

- **Radial part** ($R_{n,l}(r)$):

Example for the hydrogen atom:

$$R_{n,l}(r) = N_{n,l} \left(\frac{2r}{na_0} \right)^l e^{-\frac{r}{na_0}} L_{n-l-1}^{2l+1} \left(\frac{2r}{na_0} \right)$$

- $N_{n,l}$: Normalization constant.
- a_0 : Bohr radius.
- L_{n-l-1}^{2l+1} : Associated Laguerre polynomial.

- **Angular part** ($Y_l^m(\theta, \phi)$):

Example for $l = 1, m = 0$:

$$Y_1^0(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta$$

Illustrative Examples of n, l, m :

- **Ground State (Hydrogen Atom):**

- $n = 1, l = 0, m = 0$.
- Wavefunction:

$$\psi_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

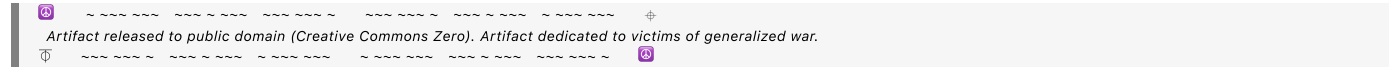
- **First Excited State ($2p$ -Orbital):**

- $n = 2, l = 1, m = 0$.
- Wavefunction:

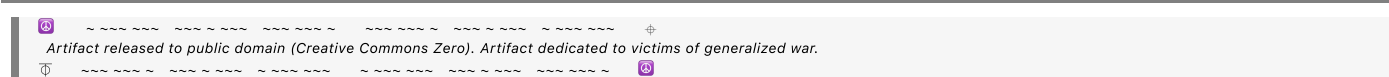
$$\psi_{2,1,0}(r,\theta,\phi) = \frac{1}{\sqrt{32\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos \theta$$

These quantum numbers define the unique states of a particle in spherical coordinates and form the backbone of quantum mechanical solutions to central potentials.

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